Supplementary Notes: Group SHAP

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Note: this topic was covered in the sixteen-week version of this course. It is provided here as optional reading and is not part of the eight-week core. Nothing in these notes is required for the weekly assignments or the course project.

SHAP (SHapley Additive exPlanations) values are a powerful tool for interpreting machine learning models by quantifying each feature's contribution to individual predictions. **Group SHAP values** extend this concept to evaluate the collective impact of feature *groups* on model performance, including testing accuracy. Here is a detailed breakdown.

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1 Group SHAP: Evaluating Feature Groups

1.1 Concept and Computation

Group SHAP values measure the combined contribution of multiple features by treating them as a single unit. This approach is particularly useful when features are logically related (e.g., demographic variables, time-series lags) and their combined effect matters more than individual impacts.

- **Mathematical Basis:** For a feature group G, its SHAP value is calculated as:

$$\phi_G \;=\; \sum_{M \subseteq F \setminus G} \frac{|M|!\,(|F| - |M| - |G|)!}{|F|!} \Big[v(M \cup G) \;-\; v(M) \Big]$$

  where $v(M)$ is the model's prediction using subset M, and F is the full feature set [3]. Under certain independence or additivity conditions, summing individual SHAP values of all features in G yields exactly ϕ_G. Otherwise, one must compute group contributions directly.

- **Tree-Based Computation:** The **Tree Group SHAP** (also known as "groupShapley") algorithm efficiently computes group contributions for tree-based models (e.g., XGBoost) by leveraging the tree structure instead of enumerating all possible feature subsets. This optimization greatly reduces computational cost [3].

1.2 Applications in Testing Accuracy

Group SHAP helps identify which feature groups drive model performance on test data:

- **Global Explanations:** Aggregate group contributions across the test dataset to rank groups by their overall impact on predictions [1, 3].
- **Bias Detection:** Analyze if specific groups (e.g., gender, age) disproportionately affect predictions, revealing potential biases in test results [2, 3].
- **Model Simplification:** Remove or merge low-impact groups to improve generalization and testing accuracy [2, 3].

1.3 Significance Testing

To statistically validate a group's impact, **Group SHAP** incorporates hypothesis testing:

- A three-cumulant chi-square approximation evaluates whether a group's contribution is statistically significant, even with skewed or sparse data [3].
- This ensures robust conclusions about a group's role in testing accuracy, avoiding false positives from random noise.

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2 Example Workflow

1. **Train a Model:** Use a tree-based algorithm (e.g., XGBoost).
2. **Compute Group SHAP:** Apply the Tree Group SHAP ("groupShapley") implementation to quantify contributions of predefined feature groups [3].
3. **Analyze Results:**
   - High positive contributions indicate groups that improve accuracy.
   - Negative contributions suggest groups that may harm performance.
   - Use significance tests to filter out noise [3].
4. **Optimize:** Refine the model by prioritizing significant groups.

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3 Comparison: Individual vs. Group SHAP

Aspect	Individual SHAP	Group SHAP (groupShapley)
Focus	Single-feature contributions	Collective group contributions
Use Case	Debugging specific predictions	Structured data, regulatory reporting
Computation Cost	Lower	Higher (but optimized via Tree Group SHAP)
Interpretability	Granular but potentially fragmented	Holistic, aligned with domain knowledge

Table 1: Comparison between Individual SHAP and Group SHAP.

In summary, **Group SHAP values** provide a structured way to evaluate how feature groups influence testing accuracy, combining interpretability with statistical rigor. This is especially valuable in domains like healthcare or finance, where feature groups (e.g., lab results, economic indicators) must be analyzed collectively.

References

[1] Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*.

[2] Molnar, C. (2021). *Interpretable Machine Learning*.

[3] Jullum, M., Redelmeier, A., & Aas, K. (2021). groupShapley: Efficient prediction explanation with Shapley values for feature groups. In *International Cross-Domain Conference for Machine Learning and Knowledge Extraction (CD-MAKE)*, Lecture Notes in Computer Science, vol. 12844, Springer.

[4] Jullum, M., Redelmeier, A., & Aas, K. (2021). groupShapley: Efficient prediction explanation with Shapley values for feature groups. arXiv:2106.12228.

[5] Aas, K., Jullum, M., & Løland, A. (2021). Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. *Artificial Intelligence*, 298, 103502. arXiv:1903.10464.